LESSON 06 · MISSION GUIDE

Monday Morning

Errors, exceptions and debugging: scripts that survive real data.

From **Mo, IT**
To **Jen**
Subject **RE: RE: RE: script fell over again**

Morning. It's Monday, so MACRO-9000 has crashed, and your export script has joined it: row 2 of this month's file says "n/a" where a salary should be. Priya forwarded me the traceback with the subject line "???". Three question marks. That's a lot for Priya.

The traceback is the most useful text Python will ever give you: what went wrong, where, in what order. Today you learn to read it, to catch the errors you expect, and to make the ones you don't expect loud instead of silent. Dave's approach was `On Error Resume Next`, which is how a grade 7 got a 40% bonus. — Mo

What you'll be able to do after today

- [ ] Read a traceback bottom-up: the error type and message, then the line that caused it
- [ ] Tell the common exception types apart: `ValueError`, `TypeError`, `KeyError`, `IndexError`, `ZeroDivisionError`, `FileNotFoundError`
- [ ] Catch an expected error with `try` / `except` and carry on, without hiding the ones you didn't expect
- [ ] Use `else` and `finally` to keep the risky line alone in the `try`
- [ ] Refuse bad input on purpose with `raise` and a message a human can read
- [ ] Find a logic bug with a well-placed `print`, or with a breakpoint in VS Code

Four big ideas

1. Errors are objects with types	A `ValueError` means "right type, wrong content" (`float("n/a")`). A `TypeError` means "wrong type altogether" (`float(None)`). Each type can be caught on its own.
2. Catch what you expect, let the rest crash	`except ValueError:` is a plan. A bare `except:` is a blindfold, the Python version of `On Error Resume Next`.
3. `try` / `except` / `else` / `finally`	Risky line in `try`, what to do if it fails in `except`, what to do if it worked in `else`, what to do no matter what in `finally`.
4. `raise` is for you	When your own code gets input that makes no sense, stop with a clear message now, rather than produce a wrong number later.

Why this matters at work

- **Real data is broken.** A script that crashes on row 2 helps nobody; one that skips row 2 *and says so* is a colleague.
- **Silent errors are the expensive ones.** A crash costs five minutes. A wrong number at the board costs a week.
- **Debugging is a skill, not luck.** Reproduce, narrow down, look at the actual values. Same method every time.

Three habits that bite

- Ten lines in one `try`: you won't know which one failed. Keep the risky line alone.
- `except Exception: pass`. Congratulations, the bug is now invisible.
- Fixing the symptom. `float("n/a")` fails because the data is wrong. Skip it *and log it*.

How today works

15 MIN
Lecture
Lars crashes scripts on purpose and reads the wreckage

60 MIN
Exercise sheet
Seven tickets, from reading a traceback to a logged safety net

15 MIN
Dave's Leftovers #6
His "robust" script. Robust like a paper umbrella

The inbox

#	FROM	TICKET	YOU'LL USE
1	Mo	Read the traceback	`IndexError` `NameError`, line numbers
2	Priya	Catch it	`try / except ValueError / else`
3	Siobhan	Which error is it?	several `except` blocks
4	Siobhan	Raise your own	`raise ValueError(...)`, `finally`, `isinstance`
5	Priya	The broken file	`except (A, B) as e`, `type(e).__name__`
6	Gerald	Print debugging	`print()` in the loop, breakpoints
7 ⭐	Mo	A safety net	a reusable `safe_float()` + a log file

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're brave), each printing the expected output
- [ ] Dave's script prints the two expected lines, one commit per bug, and your commit message names the double-counted row
- [ ] Everything is pushed to `lesson-06` and you've opened a pull request



Next lesson: The Toolbox. You keep copying the same cleaning functions into every file. Time to put them in a module of your own, borrow from Python's standard library, and install your first package with `pip`. NumPy and pandas are waiting on the other side.